

JAVA FAANG DSA HANDBOOK

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 **21 Chapters Ready (Book 1: 20 Topics + Book 2: Interview Mastery)**

PART 0 & 1: ORIENTATION & FOUNDATIONS

TOPIC 00

4 A4 PAGES

Orientation & Master Roadmap

Handbook Learning Philosophy, Target Audience, 20-Topic Sequenced Dependency Tree, 13-Block Universal Chapter Architecture & Study Schedules.

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CHAPTER 01.1**6 A4 PAGES**

Big-O & Complexity Masterclass

Algorithm definitions, Time vs Space Complexity, Asymptotic Notation (O , Ω , Θ), Growth Rate Matrix, 4 Code Analysis Rules, Recursion Call Stack & Cheat Sheet.

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PART 2: DATA STRUCTURES & CORE TECHNIQUES

TOPIC 02**10 A4 PAGES**

Arrays, Strings & Hashing (FAANG Coach)

Arrays & String Pool, HashMap Internals (Buckets, Hash Code, Treeification at 8), Frequency Arrays, Pattern Reconciliation Matrix, NeetCode Suite & Master Cheat Sheets.

[Open HTML](#)[View PDF](#)**TOPIC 03****12 A4 PAGES**

Two Pointers Masterclass

Opposite Direction, Same Direction (Reader/Writer), Dutch National Flag (3 Pointers), QuickSort Partition, Pointer Movement Visualizations, NeetCode Suite (125, 167, 26, 27, 283, 88, 15, 75, 443, 11, 42) & Master Cheat Sheets.

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Sliding Window Masterclass

Fixed Window Size K , Variable Window (Longest/Shortest), $\text{int}[128]$ Frequency Windows, At Most K to Exactly K Conversion, Window Evolution Visualizations, NeetCode Suite (121, 643, 3, 424, 567, 438, 904, 1695, 76, 992) & Master Cheat Sheets.

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TOPIC 05

10 A4 PAGES

Binary Search Masterclass

Classic BS, Boundary Search (Lower/Upper Bound), Binary Search on Answer / Solution Space, Rotated Array Search, Matrix Flattening BS, NeetCode Suite (704, 35, 374, 74, 33, 153, 875, 162, 981, 410, 4) & Master Revision Sheet.

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TOPIC 06

11 A4 PAGES

Linked List Masterclass

Java Heap Memory & References, Dummy Node Sentinel Mechanics, Fast & Slow Pointers (Floyd's Cycle), 3-Pointer Reverse Window, Before/After Pointer Animations, NeetCode Suite (206, 21, 141, 876, 234, 19, 143, 2, 138, 148, 23, 25) & Master Cheat Sheets.

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TOPIC 07

11 A4 PAGES

Stack Masterclass

LIFO Memory Model, ArrayDeque vs Deprecated Stack, Parentheses Matching, Monotonic Stack (Increasing/Decreasing), Expression RPN Evaluation, Push/Pop Visual Traces, NeetCode Suite (20, 682, 844, 150, 739, 155, 853, 22, 735, 84, 42) & Master Cheat Sheets.

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TOPIC 08

10 A4 PAGES

Queue & Deque Masterclass

FIFO Memory Model, Circular Queue Modulo Math, Deque Double-Ended Operations, Monotonic Decreasing Deque (Sliding Window Max), BFS Layer Preview, Enqueue/Dequeue Visual Traces, NeetCode Suite (232, 933, 239, 622, 649, 950, 862, 1425) & Master Cheat Sheets.

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TOPIC 09

12 A4 PAGES

Heap (Priority Queue) Masterclass

Complete Binary Tree & 0-Indexed Array Mapping, Heapify Up/Down Animations, Java PriorityQueue Comparators & Records, Top-K Min-Heap Pattern, Two-Heap Running Median, NeetCode Suite (1046, 703, 973, 347, 215, 621, 295, 658, 23, 502, 632) & Master Cheat Sheets.

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TOPIC 10

14 A4 PAGES

Trees & BST Masterclass (FLAGSHIP)

FLAGSHIP CHAPTER: Tree Terminology & Node Structure, Pre/In/Postorder DFS, BFS Level Order Queue Snapshots, Recursive Call Stack Timelines, BST Inorder Property, LCA Mechanics, NeetCode Suite (226, 104, 100, 572, 110, 102, 199, 1448, 98, 230, 235, 236, 105, 114, 124, 297, 314) & Master Cheat Sheets.

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TOPIC 11

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Trie (Prefix Tree) Masterclass

Prefix Tree Concept, Character-by-Character Insertion Visualizer, Shared Prefix Node Re-Use, Trie + DFS Grid Pruning (Word Search II), Bitwise 31-Bit Binary Trie (Max XOR), NeetCode Suite (208, 211, 648, 1268, 677, 212, 421) & Master Cheat Sheets.

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Graphs Masterclass (FLAGSHIP & WORKSHOP)

FLAGSHIP CHAPTER: Dedicated Graph Construction Workshop, Step-by-Step Adjacency List Growth Animation, 2-Page Reusable Java Interview Template Library, Multi-Source BFS, Topological Sort (Kahn's In-Degree), Disjoint Set Union (DSU), Dijkstra Shortest Path, Kruskal's MST, NeetCode Suite & Master Cheat Sheets.

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Backtracking Masterclass

State Space Search Trees, The 3-Step Choose → Explore → Undo (State Restoration) Paradigm, Subsets vs Combinations vs Permutations, 2-Page Java Template Library, Duplicate Pruning, NeetCode Suite (17, 78, 90, 39, 40, 46, 47, 131, 79, 51, 37, 212, 980) & Master Cheat Sheets.

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Dynamic Programming Masterclass (FLAGSHIP)

FLAGSHIP CHAPTER: The 8-Step 'SC-RBR-MTS' Thinking Framework, Top-Down Memoization vs Bottom-Up Tabulation, 1D/2D/Grid DP, 0/1 & Unbounded Knapsack, String Matching (LCS/Edit Distance), Interval & Tree DP, LIS $O(N \log N)$ Patience Sorting, 2-Page Template Library, NeetCode Suite & Master Revision Cheat Sheet.

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Greedy Algorithms Masterclass

Greedy Choice Property, Exchange Argument Proofs, Counter-Example Analysis (Why Greedy Fails), Sort by End Time vs Start Time, Farthest Reach Jump Windows, Circuit Deficit Net Balance, 2-Page Template Library, NeetCode Suite & Master Revision Sheet.

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TOPIC 16

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Intervals Masterclass

The 4 Canonical Interval Patterns (Merge, 3-Stage Insert, Meeting Rooms Min-Heap Overlap, Sweep Line Difference Array), Comparator Safety Rules, 2-Page Template Library, NeetCode Suite & Master Revision Sheet.

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TOPIC 17

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Bit Manipulation Masterclass

Binary Thinking & Two's Complement, The 8 Fundamental Bit Tricks, Brian Kernighan's Algorithm $n \& (n-1)$, XOR Pair Cancellation Laws, Bitmask Subset Enumeration, Bitwise Addition ('Sum of Two Integers'), 2-Page Template Library, NeetCode Suite & Master Revision Sheet.

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TOPIC 18

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Mathematics Masterclass

Euclidean GCD & LCM, Sieve of Eratosthenes $O(N \log \log N)$, Binary Exponentiation $O(\log N)$, Modular Arithmetic Laws & Fermat's Inverse, Pascal's Triangle, Cross Product Slope Checks, 2-Page Template Library, NeetCode Suite & Master Revision Sheet.

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TOPIC 19

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Advanced Data Structures Masterclass

Segment Trees with Lazy Propagation, Fenwick Trees (BIT), Sparse Tables $O(1)$ RMQ, Java `TreeMap` / `TreeSet` Navigable APIs, KMP & Rabin-Karp, Z-Algorithm, Manacher's $O(N)$ Palindromes, Tarjan's SCC & Bridges, 2-Page Template Library, Representative Suite & Senior / Staff Engineer Notes.

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🎯 BOOK 2: FAANG INTERVIEW MASTERY — SENIOR & STAFF ENGINEER PLAYBOOK

[BOOK 2 — NEW](#)[16 DENSE PAGES](#)[v5.0 FINAL](#)

FAANG Interview Mastery — Senior & Staff Engineer Playbook

The complete how-to-pass guide: Interview mindset, pattern recognition mastery, 7-step framework, Java toolkit, 5 mock interviews with full evaluation rubrics, company-specific guides for 10 companies, revision plans, and the complete FAANG Learning Framework (13 phases).

[🧠 Interview Mindset](#)[🎯 Pattern Recognition](#)[🌳 Tree Templates \(8\)](#)[🌐 Graph Templates \(11\)](#)[🔪 5 Mock Interviews](#)[🏢 10 Company Guides](#)[💰 Offer Negotiation](#)[📄 13-Phase Framework](#)[📖 Open Interactive HTML](#)[🖨️ Print as PDF](#)

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Ch 27: Company Guides (×10)

Ch 28: Revision + Study Plans

Ch 29: Playbook + Negotiation

Ch 30: FAANG Learning Framework